Linear Algebra Quiz

Duration: 45 minutes **Total marks: 30**

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Instructions. Answer all four questions. Show your reasoning and the main steps of every calculation. Work over $\mathbb{R}$ throughout. Calculators, notes, and electronic devices are not allowed.

Question 1. Geometry of an affine line **7 marks; 9 minutes**

Let $P = (1, 0, 1)$, $Q = (2, 1, 1)$, and $b = (2, 0, 0)$. The line ℓ passes through P and Q.

(a) Let $d = Q - P$ be the direction vector from P to Q. Compute d and write every point of ℓ in the form $r(t) = P + td$, where $t \in \mathbb{R}$. Is ℓ a vector subspace of $\mathbb{R}^3$? Justify your answer. **[3 marks]**

(b) Find the point p of ℓ closest to b by requiring $(b - p) \cdot d = 0$. Compute the distance from b to ℓ. **[4 marks]**

Question 2. Parameters, consistency, and null spaces **9 marks; 13 minutes**

For real parameters a and β, consider

$$A_a x = b_\beta, \qquad A_a = \begin{pmatrix} 1 & 1 & 0 \\ 0 & a & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad b_\beta = \begin{pmatrix} 1 \\ 2 \\ \beta \end{pmatrix}.$$

(a) Use row operations to classify all pairs (a, β) according to whether the system has a unique solution, no solution, or infinitely many solutions. Show the row operations that support your classification. **[4 marks]**

(b) In the case with infinitely many solutions, write the complete solution set as a particular solution plus a linear combination of null-space vectors. **[3 marks]**

(c) For $a = 0$, give a basis of $N(A_0)$ and state $\operatorname{rank} A_0$. Verify the rank–nullity relation. **[2 marks]**

Question 3. Working with an existing LU factorisation **6 marks; 8 minutes**

A matrix A has already been factored as $A = LU$, where

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

(a) A student claims that $A^{-1} = L^{-1}U^{-1}$. State the correct formula and justify it by multiplying it with A. **[2 marks]**

(b) Solve $Ax = (1, 5, 4)^T$ using forward and backward substitution. Display both intermediate systems. Do not form A or either inverse. **[4 marks]**

Question 4. Polynomial subspaces and coordinates **8 marks; 10 minutes**

Let P_2 be the vector space of real polynomials of degree at most two, and let

$$W = \{p \in P_2 : p(0) = p(1)\}.$$

(a) Show that W is a subspace of P_2. Starting with $p(t) = a + bt + ct^2$, find a basis of W and its dimension. **[3 marks]**

(b) An *ordered basis* is a basis whose vectors are listed in a specified order; coordinates are written in that same order. Verify that $B = (1, t^2 - t)$ is an ordered basis of W. The coordinate vector of q relative to B is denoted by $[q]_B = (\alpha, \beta)^T$, where $q(t) = \alpha \cdot 1 + \beta(t^2 - t)$. Find $[q]_B$ for $q(t) = 3 - 2t + 2t^2$. **[2 marks]**

(c) Add one polynomial to B to obtain a basis of P_2. Justify your choice. **[1 marks]**

(d) A different set is defined by

$$S = \{p \in P_2 : p(0)p(1) = 0\}.$$

Decide whether S is a subspace. If it is not, give two explicit polynomials that demonstrate a failure of a subspace condition. **[2 marks]**

End of quiz